Synergistic Elbow Control for a Myoelectric Transhumeral Prosthesis

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***Abstract*— This paper presents a control approach for a myoelectric transhumeral prostheses that supplements a typical two-site electromyograms (EMG) input with the inertial measurement of arm motion. The inertial measurement is employed to: 1) provide synergistic movement between the prosthetic elbow joint and intact upper arm and 2) to switch control between the myoelectric elbow and hand. In order to assess the prospective efficacy of the control method, experiments were conducted on six healthy subjects who performed a series of pick-and-place tasks within a virtual environment. The assessments compared the time required to complete the pick-and-place tasks using the proposed coordinated control approach, with the time required using a sequential control approach (i.e., the conventional approach used in commercial devices). Subjects on average completed the pick-and-place tasks 34% faster with the coordinated control approach, relative to the conventional sequential EMG method, with no difference in compensatory torso motion.**

***Index Terms*— Myoelectric control, transhumeral prosthesis, myoelectric elbow control, electromyograms (EMG) control, coordinated control.**

I. Introduction

SURVEYS of upper extremity amputees indicate that less than half regularly use their prosthetic arms [1], [2] largely due to functional limitations. Similar surveys report a desire for devices with improved functionality and more intuitive and coordinated control [3]. In order to help address such functional deficiencies, this paper presents a control approach intended to improve the functionality of myoelectric transhumeral prostheses.

A myoelectric transhumeral prosthesis typically consists of the combination of a myoelectric hand and elbow. The conventional method for control of this type of myoelectric prosthesis (i.e., the method employed in commercial devices) incorporates a two-site electromyograms (EMG) interface, in which a pair of EMG electrodes is placed over the biceps and triceps brachii muscle groups, respectively. The filtered and rectified signals measured by the agonist/antagonist pair of EMG electrodes collectively provide a single (bidirectional) control input to the prosthesis. As such, the user has only a single control input with which to control the two components (i.e., the hand and elbow) of the prosthesis. This limitation is typically addressed by multiplexing the single (bidirectional) myoelectric control signal between the myoelectric elbow joint and hand, where the multiplexing function is typically performed via a brief muscular co-contraction of the biceps and triceps muscles. Although simple and robust, this "sequential" control approach is considerably less efficient than the simultaneous joint movement characteristic of arm motions in healthy individuals. Further, the sequential method becomes increasingly inefficient with additional myoelectric degrees-of-freedom (DOFs) in the arm (e.g., powered wrist DOFs or multigrasp hands).

It is well known that arm movement in healthy individuals is characterized by the simultaneous movement of multiple joints. A control method that facilitates such simultaneous (or coordinated) movement of multiple arm joints would presumably increase the efficiency with which transhumeral amputees can perform activities of daily living (ADLs). One approach to providing coordinated control of multiple joints is to leverage known movement synergies between the (intact) shoulder and (prosthetic) elbow joints. Such synergies have been characterized in biomechanical research studies [4]–[16]. Among these, Popovic *et al.* proposed to leverage shoulder-elbow joint synergies during reaching tasks to facilitate control of functional electrical stimulation (FES) in patients with tetraplegia [6]–[11]. Kaliki *et al.* also proposed to use the measured orientation of the upper arm to predict distal arm posture for control of neuroprostheses (i.e., FES) and myoelectric transhumeral prostheses [10], [11]. Lauretti *et al.* [17], which was apparently concurrently developed with the approach presented here, proposed a coordinated controller for a transhumeral prosthesis that leverages arm joint synergies similar to those experimentally characterized in the aforementioned studies. That controller employs a pair of inertial measurement units (IMUs) on the trunk and residual limb, respectively, to provide a measure of shoulder flexion/extension and shoulder ab/adduction, and combines these measures with a two-site EMG interface, such that a sustained co-contraction of the EMG couples shoulder flexion/extension

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with elbow flexion/extension and wrist rotation with shoulder ab/adduction. In the absence of sustained co-contraction, EMG control is switched to the hand, and the elbow and wrist are immobilized.

Like the work presented in [17], this paper proposes a control method for a transhumeral myoelectric prosthesis that leverages the existence of shoulder/elbow synergies. The controller incorporates two components. The first provides a kinematic coupling between upper arm angular velocity and elbow angular velocity, the degree of which can be modulated by EMG. The second provides switching between hand control and elbow control as a function of hand velocity relative to an inertial reference frame. Unlike [17], the control approach presented here modulates the gain between the shoulder and elbow during movement using differential EMG; uses hand movement as detected by the IMU to switch between hand and elbow control rather than use sustained EMG as in [17]; and although less fundamental, employs a single IMU located on the forearm of the prosthesis (rather than two on the body).

The proposed control approach was implemented on a virtual prosthesis within a virtual environment, and its efficacy was assessed in a series of experiments on healthy subjects who conducted reaching tasks within a virtual environment, specifically comparing the ability of subjects to complete reaching tasks using the proposed controller, relative to their ability to do so using a conventional sequential controller.

II. Control Methods

A. Standard Sequential Control Approach

The conventional control approach for the control of a myoelectric transhumeral prosthesis with a myoelectric elbow and hand from a two-site EMG interface (i.e., EMG from biceps and triceps sites) is referred to herein as a sequential control approach. This controller is described here because: 1) the coordinated controller is based on the same control idea, and 2) a sequential controller is employed for comparative purposes in the assessment experiments described here. In the sequential controller, control of either the elbow or hand is selected at any given time by a short-duration co-contraction of both EMG inputs, which toggles the control to one of the two components. The component not being controlled assumes a locked state. When a component is under EMG control, the differential measurement of the two-site EMG is used to generate an angular velocity command, such that the user contracts the residual limb musculature to induce movement, and relaxes limb musculature to maintain the current configuration. Thus, if e_1 and e_2 represent the filtered and rectified EMG from the biceps and triceps muscles, respectively, the angular velocity reference (or command) for the selected, or active, component is given by:

$$\omega_{emg} = \begin{cases} k_1 e_1 & if e_1 \geq e_2 \\ -k_2 e_2 & otherwisee \end{cases} \tag{1}$$

where k_1 and k_2 are normalizing gains, typically chosen to provide nominal symmetry of movement (i.e., similar angular velocity responses) in response to similar contractile efforts.

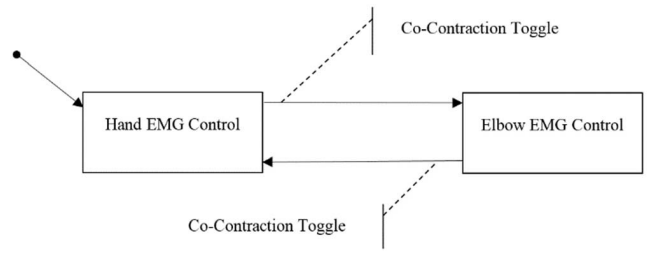

Fig. 1. FSM of sequential elbow/hand control structure.

Typical signal conditioning such as hysteresis and a dead band around an EMG noise floor are incorporated to facilitate well-behaved, deliberate control. A finite state machine (FSM) depicting the sequential control structure for a typical transhumeral prosthesis is shown in Fig. 1.

B. Coordinated Controller

The proposed coordinated controller supplements the standard two-site EMG interface with a single-axis gyroscope. Note that a gyroscope mounted on the forearm of a prosthesis and oriented to measure angular velocity parallel to the elbow axis of rotation can when combined with measurement of the prosthetic elbow angular velocity, provide real-time measurement of the component of upper arm angular velocity collinear with the elbow axis of rotation. Given this measurement (of upper arm angular velocity collinear with the elbow), the coordinated controller is a straightforward modification of the standard elbow controller. Specifically, the EMG component of the commanded elbow angular velocity is proportional to the differential EMG measurement, as given in (1). The synergistic component (i.e., reaching component) of the elbow angular velocity is proportional to the measured shoulder angular velocity:

$$\omega_{sy} = k_3 \omega_{ua} \tag{2}$$

where ω_{sy} is the synergistic component of the elbow angular velocity command, k_3 is a coupling coefficient, and ω_{ua} is the measured upper arm angular velocity. The commanded elbow angular velocity is the superposition of the EMG and synergy components of angular velocity, given by:

$$\omega_{eb} = \omega_{sy} + k_4 \omega_{emg} \tag{3}$$

where ω_{eb} is the superposed elbow angular velocity command and k_4 is a weighting factor between the two components of elbow angular velocity command. Thus, the coordinated controller is in implementation a relatively simple modification of the conventional EMG elbow controller. Also, the coordinated elbow controller (3) behaves similar to the standard elbow controller (1). Specifically, both controllers use EMG in the same manner to command elbow angular velocity, although in the case of the coordinated controller, the default behavior is one of kinematic coupling between elbow and upper arm angular velocity, while in the case of the standard controller, the default is an absence of arm/elbow coupling.

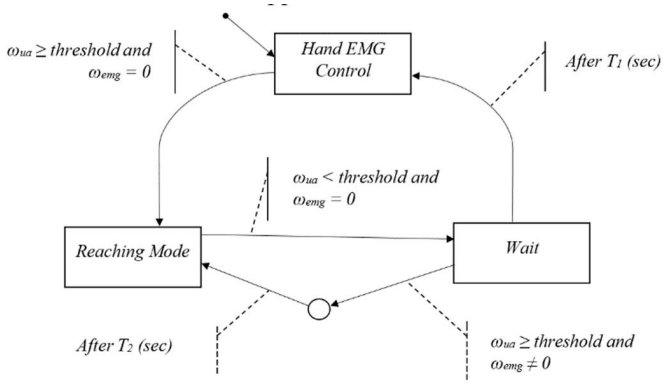

Fig. 2. FSM of coordinated elbow/hand control structure.

Like the standard sequential controller, the coordinated controller is intended to control only one myoelectric component (i.e., elbow or hand) at any given time, and thus the control must similarly be switched between the elbow and hand component. When the hand control is active, the prosthesis utilizes the standard proportional EMG control (i.e., equation (1)). When the elbow control is active, the prosthesis utilizes the coordinated control equation (i.e., equation (3)). Rather than use co-contraction to switch between the two controllers, the coordinated controller is further modified (relative to the standard sequential control) to leverage the presence of the gyroscope to facilitate the process of switching between hand and elbow control. Specifically, when the elbow velocity (3) remains essentially zero for a short duration, and when both components on the right-hand side of (3) also remain essentially zero for this duration (i.e., both the upper arm angular velocity and EMG command are zero), the prosthesis control is automatically switched to the hand. Note that this condition implies that the hand is essentially remaining fixed relative to the user's body, such as would be the case if a user were about to grasp a stationary object. When in the hand control state, and when the upper arm angular velocity becomes essentially non-zero (i.e., the upper arm begins to move in the plane orthogonal to elbow rotation), the EMG control automatically switches back to the elbow. In this manner, all elbow movement must by initiated by upper arm (i.e., shoulder) movement. The use of this rubric removes the need to use co-contraction to toggle control between components. As such, co-contraction can either be eliminated from the control structure, or can be preserved for use for the control of other components. Fig. 2 shows a FSM of the coordinated controller, depicting the switching structure of the coordinated control approach.

III. Experimental Methods

The relative performance of the coordinated controller was assessed and compared with a standard sequential EMG control approach in a set of experiments in which six healthy subjects performed several different pick-and-place tasks within a virtual environment. The virtual environment enabled both a well-controlled experimental environment, and a well-controlled assessment of performance.

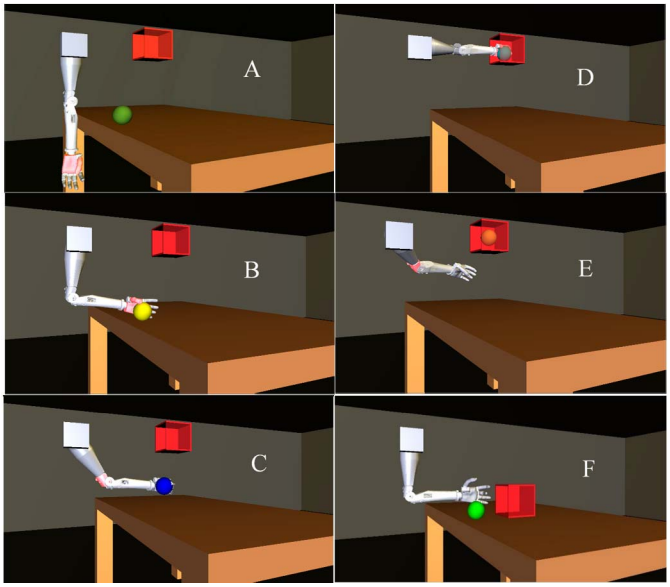

Fig. 3. The virtual environment used in the controller assessments. (A) The elbow joint starts each task fully extended, and must be controlled by the user to configure the elbow to pick the ball. (B) The hand starts fully opened; the ball changes color from green to yellow when the hand is positioned to grasp it; the hand must be closed by the user around the ball in order to grasp it; the ball changes color from yellow to blue to indicate that it has been grasped. (C) Once grasped, control is switched to the elbow to perform the reaching motion toward the target box. (D) The ball must be positioned in the middle of the box, after which it changes color to light blue, to indicate that it is ready to be released. Control must be switched to the hand to release the ball. (E) After successful release, the ball changes color to orange, the completion time is recorded, and (F) a new ball and box appear for the next reaching task. The tasks were performed continuously, such that the timer for the next task started when the ball was released from the previous task.

A. Virtual Environment and Task

The virtual assessments consisted of a series of pick-and-place tasks in which each subject was required to pick up a ball resting on a table and place the ball into a target box, which was randomly positioned in one of twelve target locations. The twelve target locations corresponded to a spatial matrix of box locations, configured to be two boxes across, three boxes high, and two boxes deep. Each box was open in the front and approximately cubic, with a characteristic dimension approximately twice the ball diameter. A representative pick-and-place task is depicted in the sequence of screen shots shown in Fig. 3, where the activity associated with each frame is described in the caption.

The virtual environment in which the assessments were performed included an 8-DOF virtual arm, where the 8-DOFs included: 1) the three-dimensional location of the shoulder joint within the virtual space (3 DOFs); 2) the angular orientation of the upper arm in the virtual space (3 DOFs); 3) the elbow angle (1 DOF); and 4), the hand configuration, which in this case was a single-DOF device, the grasp of which was characterized by a single angle and corresponding angular velocity (1 DOF). Instrumentation on each subject provided real-time input that defined motion of the upper arm and shoulder DOFs, such that the upper arm and shoulder tracked in real-time the physical motion of each subject. Motion of the elbow and hand DOFs, however, were based on the

(real-time) output of either the sequential or coordinated arm controllers, depending on which controller was being used at any given time. These controllers were driven by real-time EMG measurements from the user's healthy arm, and in the case of the coordinated controller, an IMU affixed to the subject's arm to measure upper arm angular velocity. Thus, the virtual arm emulated a transhumeral prosthesis, in which the torso and upper arm movement tracked in real-time the corresponding movement of the seated subject, while both the hand and elbow components of the prosthesis were controlled via real-time EMG measured from the subject (in the case of the sequential controller), or by both real-time EMG and IMU measurement (in the case of the coordinated controller).

B. Assessment Protocol

Six healthy, non-amputee subjects participated in the experiments, which were approved by the Vanderbilt

University Institutional Review Board. Each subject participated in five trials for each controller where, each trial consisted of 36 virtual pick-and-place tasks, where each of the 12 target locations appeared three times in a randomized order. All subjects were seated in a chair in front of the computer monitor, which remained in a fixed location once the trial began. Subjects were required to use their left arm in the experiments, since both the instrumentation and virtual prosthesis model were configured for the left arm. The trials for each subject were conducted over a one-week period. In order to account for order effects, half of the subjects used the sequential controller first, followed by the coordinated controller, while the other half of subjects performed the trials in the reverse order. Note that each subject was required to complete each pick-and-place task. If a virtual ball was dropped prior to placement in the box, the ball would be automatically reset in the starting position for the subject to re-attempt the task, until the task was successfully completed. Once the task was successfully completed, the ball was reset in the starting position, and another target location (i.e., box location) was randomly presented.

For each subject, an initial session was used to establish EMG thresholds and normalization gains, followed by an introduction to the control method, and a practice period during which the subject could practice controlling the elbow and hand, and could practice performance of the pick-and-place tasks. The duration of the training/practice phase varied by subject, but typically lasted between 15 and 30 minutes, as per each subject's preference. Once sufficiently practiced, the subject completed a series of five trials using the assigned controller. Each subject required approximately four days to complete the training and experiments. The experiments consisted of a total of ten trials (five for each controller type). Each trial entailed 36 pick-and-place tasks, so each subject completed a total of 360 pick-and-place tasks over the four-day period, or on average 80 pick-and-place tasks per day.

C. Controller Thresholds and Gains

The two controllers (equations (1) and (3)) collectively employ four gains, which were established for each subject prior to performing the virtual assessments. Note that once these gains were established, they were maintained in the controller for each subject throughout all experimental sessions (i.e., for the several-day duration of the each subject's involvement in these experiments). The gains k_1 and k_2 ensure that the maximum EMG for each subject and each EMG channel corresponds to a maximum elbow angular velocity. These were established in pre-trial experiments for each subject, in which several maximum voluntary contractions were performed, and the respective gains were established by correlating the mean EMG corresponding to the contraction with the maximum commanded elbow angular velocity (set at 50 deg/s). The coordinated controller also employs two additional gains, k_3 (which determines the degree of kinematic coupling between the shoulder and elbow) and k_4 (which determines the relative weighting of EMG command to IMU command in the controller. For all trials and all subjects, both these gains were set to one, such that in the absence of EMG, elbow angular velocity would exactly equal upper arm angular velocity, while in the absence of upper arm motion, the coordinated controller reduced exactly to the sequential EMG controller.

D. Instrumentation and Apparatus

In order to perform the assessments, each subject was instrumented with sensors to provide real-time input to drive the virtual prosthesis. Measurement of torso motion was also used to characterize movement of the torso for purposes of performance assessment. In total, the experimental instrumentation consisted of: 1) a pair of EMG electrodes; 2) an elbow immobilizer; 3) an upper arm-mounted IMU; and 4) a trunk-mounted IMU. The trunk IMU was not used for prosthesis controller input, but rather was used to 1) provide Cartesian motion of the trunk (i.e., shoulder joint) relative to the task frame to drive the virtual torso motion and 2) to characterize compensatory movement of the trunk. The upper arm IMU was used to provide upper arm motion in the task frame, in order to drive the movement of the virtual upper arm, and also as an input to control the prosthesis for the coordinated control approach. Since movement of the intact elbow, wrist, and hand were not used in the experiments, a hand, wrist, and elbow joint immobilizer was employed to preclude movement of the (non-virtual) joints during the experiments and better represent the muscular conditions in the residual limb of an amputee subject. Use of the immobilizer, however, impinged on the upper arm EMG electrodes and obstructed upper arm EMG measurement. In order to provide unobstructed EMG input for the prosthesis controller(s), the EMG electrodes were moved to the forearm, where they were free of impingement by the immobilizer. Although such placement is less representative of the musculature employed by transhumeral amputees, since both controllers employed the same EMG sites in the same manner (i.e., equation 1), any substantial effect from this modification should wash out of the comparison, thus indicating the relative value of superimposing the synergistic component of control on top of the EMG component.

1) EMG Measurement and Signal Conditioning: EMG data was acquired using two disposable conductive gel Ag/AgCl

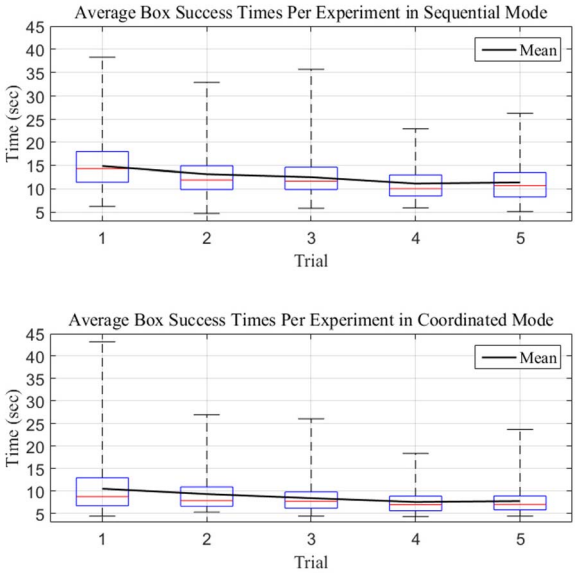

Fig. 4. Distribution of task completion times averaged across all subjects for each of the five trials, for each controller type.

electrodes (Norotrode 20, Myotronics, Inc.) positioned over the subject's anterior and posterior surfaces of the left forearm with a reference electrode positioned proximal to the elbow. The acquired raw EMG signals of the electrodes were preamplified with a gain of 100 and digitally sampled using an EMG processing chip (Texas Instruments ADS1298), which also provided bandpass filtering (between 20 Hz and 500 Hz) and rectification. The rectified signal was sent via serial peripheral interface (SPI) to a PIC32 microcontroller (Microchip, Inc.), where it was low-pass-filtered at 5 Hz to provide an envelope of spatially integrated EMG activity. The (two) processed EMG signals were then streamed from the PIC32 via a controller area network (CAN) bus to the real-time environment provided by Simulink Real-Time Workshop (RTW).

2) Shoulder/Arm Angular Velocity Measurement: An IMU was incorporated on the same circuit board used for EMG data acquisition and signal processing, which was affixed to the arm immobilizer affixed to the subject's upper arm. The IMU used for upper arm motion was a high-performance 9-axis unit (VectorNav Technologies VN-100 SMD), which was used to both drive the upper arm orientation within the virtual environment (for both controllers), and also to provide the angular velocity command for the coordinated controller.

3) Torso Motion Measurement: The torso motion was measured using a separate 6-axis IMU (Invensense MPU-6555) mounted to a separate circuit board via an elastic band to the center of each subject's chest, approximately at the xiphoid process. The torso IMU provided real-time torso orientation in both the sagittal and coronal planes, which was used to: 1) drive the torso motion within the virtual environment, and 2) provide a measure of compensatory motion during the experiments.

IV. Results

Figure 4 shows the distribution of task completion times averaged across all subjects for each of the first five trials, for each controller type, and also shows the mean completion time (for each trial) across all six subjects (which is the black line connecting all trials). The numerical values of the median completion times corresponding to each trial, in addition to the interquartile ranges for each, are enumerated in Table I. The data in Table I (and Fig. 4) indicate learning trend in both controllers. A Kolmogorov-Smirnov test indicated the data were non-normal, and as such Wilcoxon signed rank tests with $p < 0.01$ were used to assess significance in differences between medians associated with each controller type. Further, since data was found to be non-normal, the extent of spread about the median value is indicated in the results via interquartile ranges (IQRs) rather than by standard deviations.

TABLE I
Average Task Completion Times

Trial	Sequential Completion Times (s)	Coordinated Completion Times (s)
	median (IQR)	*median (IQR)*
1	14.3 (6.5)	8.8 (6.3)
2	11.9 (5.1)	7.9 (4.3)
3	11.7 (4.8)	7.7 (3.7)
4	10.1 (4.6)	7.0 (3.3)
5	10.7 (5.3)	7.0 (3.1)
Median trials 4-5	**10.6 (4.9)**	**7.0 (3.1)**

In order to reduce the effect of learning, performance data was only considered after performance in both controllers had converged, which was assessed by differences in median values between trials, specifically when differences in medians between trials (as indicated by a Wilcoxon signed rank test) were no longer significant. Based on this criterion, only the last two trials (i.e., trials four and five) were included in the performance assessment. As given in Table I, the median task completion time for the sequential controller (across all subjects and across trials four and five) was 10.6 s, while the average task completion time for the coordinated controller (also across all subjects and trials four and five) was 7.0 s (with differences in medians significant). As such, the coordinated controller enabled completion of the pick-and-place tasks on average 34.0% faster than the sequential controller. Further, this reduction in completion times was consistent across all box (i.e., reaching) locations, as indicated in Fig. 5, which shows the median completion times (and IQRs) across all subjects for each box location, for trials four and five (i.e., each bar represents the median of 36 trials). Differences in medians for each box location were significant. As such, the proposed method provided a consistent performance advantage, relative to the sequential approach, for all reaching movements represented in this study (i.e., regardless of specific reaching location).

In order to ensure that decreases in completion times were not a result of increases in compensatory (i.e., torso) motion, torso angles with respect to the vertical in the sagittal and frontal planes were recorded at the time each reaching task

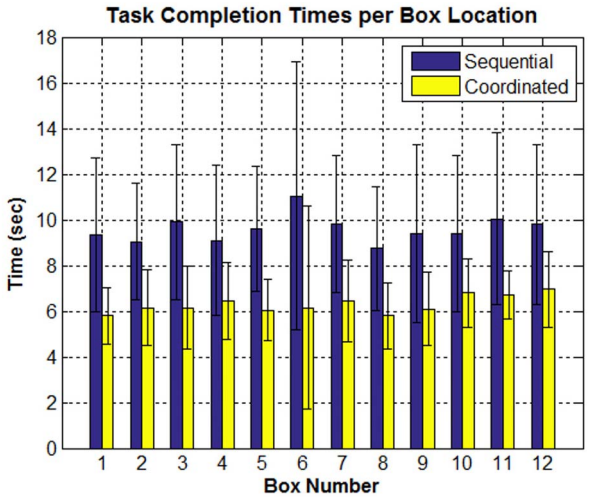

Fig. 5. Median task completion times across all subjects and trials, corresponding to each respective target box location, representing a variety of reaching movements. The error bars indicate plus/minus one IQR.

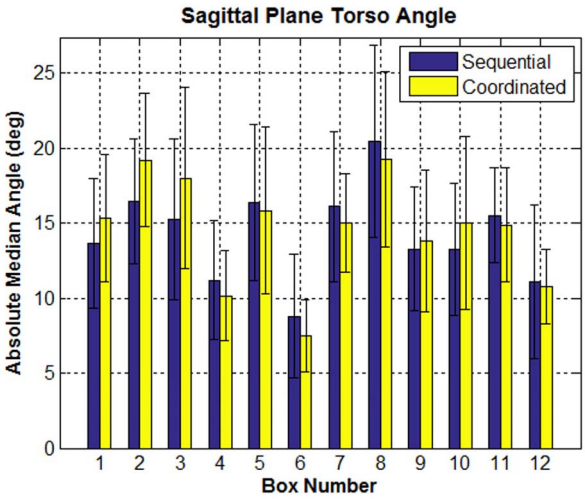

Fig. 6. Comparison of the median forward lean angle employed to reach each target location. Error bars represent plus/minus one IQR.

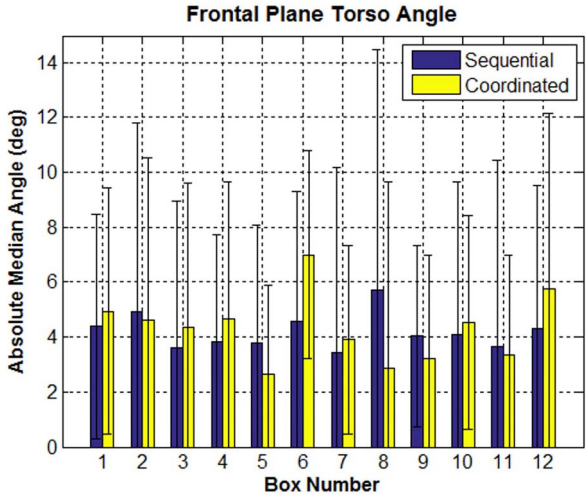

Fig. 7. Comparison of the median sideways lean angle employed to reach each target location. Error bars represented plus/minus one IQR.

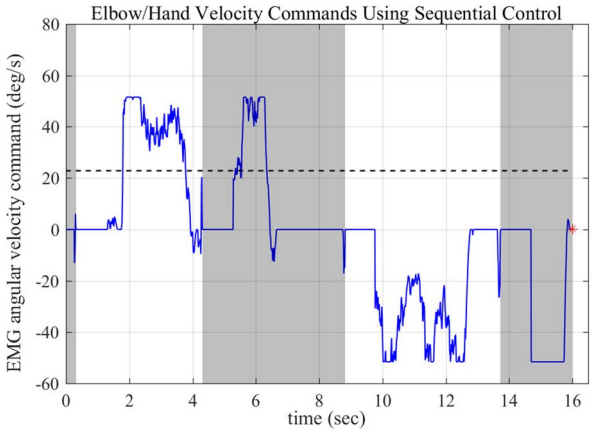

Fig. 8. Arm (elbow and hand) velocity commands during a representative pick-and-place task (box no. 3) performed with the sequential EMG controller. The gray bands indicate the periods of hand control, while the white bands indicate the periods of elbow control. Positive and negative elbow commands indicate elbow flexion and extension respectively.

was completed (i.e., when the ball was successfully released in the target box). These measurements in effect characterized the torso configuration upon completion of each reaching task, and specifically how much forward (Fig. 6) and sideways (Fig. 7) lean was employed to reach the target location. Although slight differences exist in the medians, Wilcoxon tests indicated in all cases (i.e., all box locations) the average compensatory torso motions were not statistically different between the two controllers. As such, median task completion times across all subjects and locations were 34.0% faster with the coordinated controller, with no significant difference in compensatory torso motion.

V. Discussion

Based on the virtual experiments, use of an EMG-modulated shoulder/elbow synergy, combined with an IMU-based approach to toggling between active components, appears to provide a significant reduction in task completion times for pick-and-place reaching tasks, without altering the extent of corresponding torso motion. As previously mentioned, the proposed elbow control approach can be regarded as one quite similar to a standard EMG control approach, but rather than use EMG to command elbow velocity about an operating point described by fully decoupled upper and lower arm movements, the proposed controller uses EMG to command elbow velocity about an operating point that assumes a simple kinematic coupling between the limb segments.

A. Movement Strategies

Controller data during performance of a representative pick-and-place task (i.e., while picking and placing the ball into box number 3) using the sequential EMG controller (Fig. 8) and the coordinated controller (Fig. 9), respectively. Specifically, Fig. 8 shows the EMG command for the sequential controller over the task performance period, where the gray bands

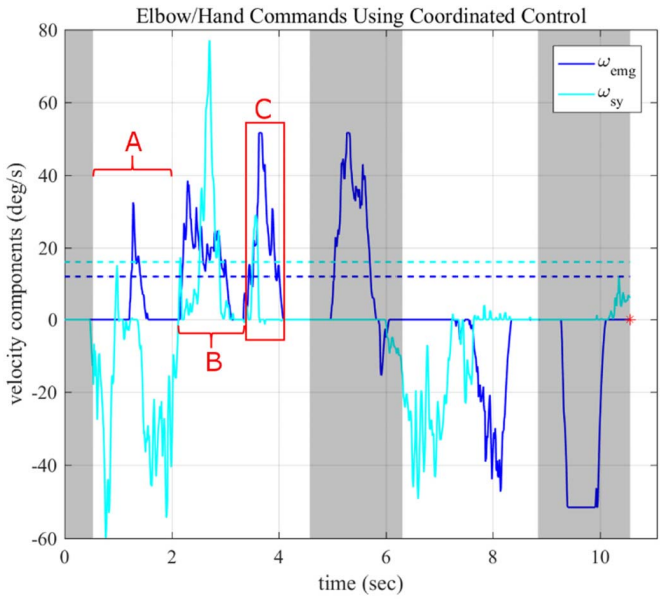

Fig. 9. EMG (dark trace) and IMU (light trace) elbow angular velocity commands during a representative pick-and-place task (box no. 3) performed with the coordinated controller, and corresponding RMS averages over the task duration (dashed lines). The gray bands indicate the periods of hand control, while the white bands indicate the periods of elbow control. Positive elbow angular velocity commands indicate flexion while negative commands indicate extension.

indicate periods of hand control and the white bands indicate periods of elbow control. The periods of activity correspond to positioning the elbow to pick up the ball, grasping the ball, extending the elbow (and flexing the shoulder) to place the ball, and releasing the ball. One can note the co-contraction "spikes" in the EMG signals that separate the period of hand and elbow control.

Figure 9 shows the corresponding controller data, for the same subject, during the same pick-and-place task (i.e., box number 3), while using the coordinated controller (i.e., equation 3), showing both the IMU component of elbow angular velocity (equation 2) and the EMG component (equation 1), over the entire task performance period. Since the weighting gains are unity, the total elbow angular velocity command (equation 3) is the direct sum of these components. As in Fig. 8, the gray bands indicate periods of hand control and the white bands indicate periods of elbow control. Note that co-contraction is not present at the transitions between elbow and hand control, since it is not used in the coordinated controller.

As previously described, the initial configuration of the arm is in a vertical orientation with fully extended elbow (i.e., the arm is hanging by the user's side). The sequence of commands shown in Fig. 9 is as follows. Recall that the state controller resides in hand control when the hand remains motionless, and as such starts in the hand control state. In order to switch to the elbow control state, the user must move his shoulder (see Fig. 2) in order to enter the elbow control state. This movement is shown in the period noted as "A" in the figure, during which the user flexes his shoulder slightly, which switches control to the elbow. Note that this initial shoulder flexion would normally result in elbow extension; however, since the elbow starts fully extended, it cannot extend further, and thus the initial elbow extension command seen in period "A" switches the arm into elbow control, but does not result elbow extension. In period "B," the shoulder is extended (to return it to the neutral position), which results in elbow flexion, which is simultaneously supplemented with an EMG flexion command to increase the rate of elbow flexion. In period "C," the shoulder has reached its desired position, but the user continues elbow flexion via an EMG flexion command. Thus, period "B" illustrates simultaneous usage of the IMU and EMG inputs to the flex the elbow, while period "C" illustrates subsequent adjustment of the elbow position with EMG alone. Following the initial configuration of the shoulder and elbow to grasp the ball, a brief period of inactivity switches the controller to hand control (between approximately 4.5 and 6 s), where EMG is used to close the hand and grasp the ball. From approximately 6.5 to 7.5 s, the shoulder is flexed, which switches the arm to elbow control, and also extends the elbow as a function of shoulder movement. Between approximately 7.5 and 8.5 s, the shoulder has reached its final configuration (i.e., has fully flexed), although the elbow requires additional extension, which is performed with EMG alone. Once both the shoulder and elbow joints are in their final configuration (at approximately 9 s), the momentary lack of shoulder and elbow movement switches the controller to hand control, and the user uses EMG to release the ball.

Figure 9 also shows the root-mean-square (RMS) values of the respective velocity commands (dashed lines). For this task, the average RMS angular velocity command resulting from the IMU component was 14.0 deg/s, while the RMS command resulting from the EMG component was 12.0 deg/s. As such, for this representative task, the majority (~60%) of elbow movement resulted from the IMU component. Further, the average speed of elbow movement across the trial was 28.0 deg/s (the sum of the components), which is approximately 27% faster than the corresponding average speed with the sequential control approach (which was 22.8 deg/s).

B. Subjective Feedback

Although subjective feedback has clear limitations, following the trials, subjects indicated that the coordinated controller provided more intuitive elbow movement that was easier to use than the EMG-only (sequential) elbow controller. Subjects also reported that switching between hand and elbow control was considerably easier without the need for co-contraction.

C. Assessment Limitations

Limitations of this assessment include: 1) use of healthy rather than amputee subjects; 2) use of a virtual rather a physical environment; 3) use of a virtual rather than physical prosthesis; 4) use of forearm rather than upper arm EMG; and 5) only reaching tasks were characterized. Another limitation of this study is that the data does not explicitly indicate the relative functional value provided by the IMU component of the elbow controller, versus the value of using the IMU to

perform switching of control between the hand and elbow components.

D. Potential to Include Wrist Control

Although a myoelectric wrist was not employed in this study, the control method could potentially be used with one. Since EMG co-contraction is not required for switching between control of the elbow and hand with the proposed approach, such co-contraction can be reserved for switching between control of a myoelectric hand and wrist, rather than between an elbow, hand, and wrist. Alternatively, although beyond the scope of this paper, the method proposed here could be expanded to exploit IMU information to facilitate improved wrist control in a transhumeral prosthesis with elbow, wrist, and hand components. For example, the authors recently investigated a method that employs IMU information for wrist control in a transradial prosthesis, specifically by using abduction and adduction movements of the upper arm to control wrist rotation [18]. Such a method could be combined with the method described here; the authors have not investigated this combined approach, however, and as such the prospective efficacy of combining these methods is unclear.

E. IMUs and Myoelectric Prostheses

One of the significant challenges in providing highly-functional myoelectric prostheses is the limited communication channel by which a user can communicate control and intent with the prosthesis. Traditionally, the information included in this communication channel has been solely EMG information. Further, the communication channel has traditionally been limited to two EMG sites, particularly in the case of transhumeral prostheses. EMG measurement, although critically important to the control of such prostheses, is also sensitive to noise, and somewhat non-stationary due to skin conditions (e.g., perspiration) and movement between the socket and skin. IMUs are now widely available; have relatively low cost (e.g., current 6-axis IMUs are available for approximately 5 USD per IMU); and are quite small (e.g., on the order of 10 mm^2). These sensors provide reasonably high-bandwidth (i.e., on the order of several Hz) kinematic measurement of the movement of the residual limb (i.e., the upper arm in the case of a transhumeral prosthesis); they generally provide low noise measurement (relative to typical EMG); and are generally not affected by skin condition. Since the residual limb is under direct control of the central nervous system (CNS), an IMU therefore contains indirect measure of neural activity (i.e., it encodes neural commands associated with arm movement). Further, a user generally retains full proprioceptive sensation of the intact shoulder joint, and therefore is generally capable of high-fidelity control of the shoulder joint. In contrast, EMG can be difficult to control without visual feedback. For example, in the absence of visual feedback, a user can generally maintain a given shoulder position over time, but is less able to maintain a given EMG amplitude. As such, although IMU information will certainly not replace EMG information, it can be integrated with EMG measurement in a number of ways to increase the amount of information carried in the communication channel between user and prosthesis, and therefore to potentially improve the efficacy and efficiency of prosthesis control.

Among the potential methods of integrating IMU and EMG information, as investigated in this paper, IMU measurement can be employed to provide an inter-joint kinematic coupling and/or to govern switching of EMG control between components. In a different approach, IMU information could be used to supplement a vector of EMG measurements for use in a pattern recognition methodology. A recent example of this approach is reported in [19], in which the authors enhance EMG information with IMU measurement to improve grasp pattern recognition for control of a myoelectric multigrasp hand. In another recent example of incorporating IMU to improve myoelectric control, researchers describe the use of IMU information to provide activity context via a finite state machine, where the finite state is used to select control parameters associated with a myoelectric controller [20]. In addition to these approaches, IMU information could be used as a control input distinct from EMG, in a manner similar to a body-powered prosthesis (although using shoulder movement rather cable excursion from a shoulder harness). Regardless of the specific manner in which IMU information is exploited, the authors believe such information has the potential to enable a user improved control of a myoelectric prosthetic arm.

VI. Conclusion

This paper proposed a control method for the control of a transhumeral prosthesis with myoelectric hand and elbow components. The control method supplements the standard two-site EMG control with an IMU, which provides 1) kinematic coupling between the (prosthetic) elbow and (intact) upper arm, and 2) automatic switching between hand and elbow control, based also on movement of the intact shoulder. The authors conducted an assessment of the coordinated control approach on six healthy subjects in a virtual environment involving a series of pick-and-place tasks, where location of the object placement was varied widely. Results from this assessment indicated that the median task completion times were in aggregate 34% less when using the coordinated controller, relative to completion with a traditional sequential transhumeral myoelectric control approach, without affecting the extent of compensatory torso movement. In future work, the authors plan to conduct these assessments with amputee subjects using a myoelectric transhumeral prosthesis.

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